Linux Group Administration

Concepts, core commands with examples, command chart, and interview Q&A

Privilege note: commands are written without `sudo`, as asked - that does **not** make them unprivileged.
ROOT = changes the system; run from a root shell (or as the group administrator where noted).
ANY USER = read-only, safe for everyone. Replace words in `<angle brackets>` with real names.

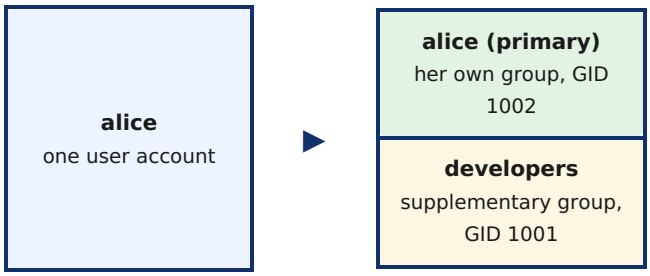
1. What is a group?

A **group** is a named collection of user accounts that share the same access permissions to files and other resources. The admin sets permissions **once for the group** instead of once per user.

Analogy: instead of telling the librarian "Asha, Ravi and Meena may enter the lab", you say "**Class 3A** may enter the lab". Anyone added to Class 3A gets the access automatically - and removing one name from the class list removes the access, without touching the lab door rules.

Why groups are used

Simpler permission management - add or remove one person and their access follows the group.
Better security - access is limited to exactly the right people.
Tidy organization - users are collected by team or role (developers, testers, finance).



Primary vs supplementary groups

Every user has **exactly one primary group** - stored in `/etc/passwd`; it is the default group for new files they create. A user can also belong to **any number of supplementary groups** - the extra memberships listed in `/etc/group`.

Correction to the slide: it says a group name can be 255 characters. The shadow-utils `groupadd` man page documents up to **32 characters**; the exact rules depend on the distribution and tool version, not a universal limit. Keep names short.

2. The two group files

Group information lives in two files:

/etc/group

Group name, password placeholder (x), GID, member list.

Readable by everyone.

/etc/gshadow

Hashed group password, group administrators, members.

Readable by root only.

Fields of /etc/group

```
developers:x:1001:alice,bob
```

developers	x	1001	alice,bob
group name	**password placeholder**	**GID**	**members**

Fields of /etc/gshadow

```
developers:!:alice:bob
```

developers	!	alice	bob
group name	**password**	**group admins**	**members**

A password field of `!` means **nobody can join with a password** - existing members are unaffected. Here `alice` is the group administrator; note that **an admin is not automatically a member**. The members field lists only supplementary members, so a user whose *primary* group is `developers` may not appear there - use `id <user>` for the full picture.

3. Core commands with examples

Create a group ROOT

```
groupadd developers          # create the group
groupadd -g 2000 testers   # choose the GID yourself
getent group developers    # check it worked
```

Add a user to a group ROOT

Correction to the slide: it shows `usermod -a <group> <user>`, which is invalid. `-a` works only together with `-G`, and the group name is the value of `-G`.

```
usermod -a -G developers alice     # add alice, KEEP her other groups
usermod -a -G dev,testers alice    # add to several groups at once
gpasswd -a bob developers          # alternative: ROOT or the group admin
```

`-a` means append. Without it, `-G` **replaces** the user's whole supplementary group list - any group not named is removed (the primary group is untouched). Change the primary group with `usermod -g developers alice` ROOT. A logged-in user keeps the old group list until they log out and in again (or run `newgrp`).

Give a user administration access to a group ROOT

```
gpasswd -A alice developers   # make alice the group administrator
```

A group administrator can then add and remove members of that group without being root (`gpasswd -a bob developers` / `gpasswd -d bob developers`). **Careful:** `-A` replaces the whole admin list; give several admins comma-separated. Being admin does not make alice a member - add her with `gpasswd -a` if she also needs the group's access.

Remove a user from a group ROOT or group admin

```
gpasswd -d alice developers   # removes alice from developers only
```

Her other groups stay; running sessions keep the old access until she logs out. This edits only member lists - it cannot change a user's primary group (use `usermod -g`).

Delete a group ROOT

```
groupdel developers
```

`groupdel` refuses if the group is some user's primary group - change those users first. Files owned by the group are **not** deleted; they keep the old GID and show a number in `ls -l` until you `chgrp` them.

Rename / renumber a group ROOT (extra)

```
groupmod -n devteam developers   # rename
groupmod -g 3000 devteam         # change GID
```

Check group information ANY USER

```
groups alice              # groups alice belongs to
id alice                  # UID, primary GID and ALL groups
getent group developers   # one group's entry from /etc/group
```

Sample output of `id alice`:

```
uid=1002(alice) gid=1002(alice) groups=1002(alice),1001(developers)
```

Primary group `alice`, plus supplementary group `developers`.

Why new access needs a fresh login



Group membership is read at login time - a running session does not pick up changes.

4. Command reference chart

Command	Use	Needs
`groupadd <g>`	Create a group	ROOT
`groupadd -g <gid> <g>`	Create with chosen GID	ROOT
`usermod -a -G <g> <u>`	Add user to group, keep other groups	ROOT
`usermod -G <g1,g2> <u>`	REPLACE all supplementary groups (needs -a to append)	ROOT
`usermod -g <g> <u>`	Set primary group	ROOT
`gpasswd -a <u> <g>`	Add one user to a group	ROOT / group admin
`gpasswd -d <u> <g>`	Remove user from a group	ROOT / group admin
`gpasswd -A <u> <g>`	Set group administrator(s)	ROOT
`gpasswd -M u1,u2 <g>`	Replace the full member list	ROOT
`gpasswd <g>`	Set group password	ROOT / group admin
`gpasswd -r <g>`	Remove group password (only members can then newgrp)	ROOT / group admin
`gpasswd -R <g>`	Restrict group (password set to !, no password joins)	ROOT / group admin
`groupmod -n <new> <old>`	Rename a group	ROOT
`groupmod -g <gid> <g>`	Change GID	ROOT
`groupdel <g>`	Delete a group	ROOT
`groups [user]`	Show groups of a user	ANY
`id [user]`	Show UID, GID and all groups	ANY
`getent group [g]`	Show group entries	ANY
`cat /etc/group`	View the group file	ANY
`cat /etc/gshadow`	View the group shadow file	ROOT
`newgrp <g>`	Shell with <g> as active group (members: no password)	ANY (member)

5. Interview questions with model answers

Q1. What is a group in Linux?

A named collection of user accounts that share the same access permissions. Admins set permissions once for the group instead of per user.

Q2. Why do we use groups?

Simpler permission management, better security by limiting access to the right people, and users organized by team or role.

Q3. Which files store group information?

`/etc/group` (name, GID, members; readable by everyone) and `/etc/gshadow` (passwords, admins, members; root only).

Q4. Explain the fields of /etc/group.

Four colon-separated fields: group name, password placeholder (x), GID, comma-separated members.

Q5. Explain the fields of /etc/gshadow.

Group name, encrypted password, group administrators, group members. A `!` password means nobody can join using a password.

Q6. How do you create a group?

`groupadd <name>`, as root. Example: `groupadd developers`.

Q7. How do you add a user to a group without removing their existing groups?

`usermod -a -G <group> <user>`. The `-a` (append) is essential; `-G` alone replaces all supplementary groups.

Q8. How do you make a user a group administrator?

`gpasswd -A <user> <group>`. They can then add and remove members of that group without root rights.

Q9. How do you remove a user from a group?

`gpasswd -d <user> <group>`.

Q10. How do you delete a group?

`groupdel <group>`. It fails if the group is a user's primary group; files owned by the group are not deleted.

Q11. Difference between primary and supplementary group?

Primary: stored in `/etc/passwd`, default group of new files. Supplementary: extra memberships listed in `/etc/group`.

Q12. How do you check which groups a user belongs to?

`groups <user>` or `id <user>`.

Q13. Why does a user not get new group access immediately?

Membership is read at login. They must log out and back in, or start a shell with `newgrp`.

Q14. What is the difference between usermod -g and usermod -G?

Small `g` sets the primary group. Capital `G` sets the supplementary group list.

Q15. Is a group administrator automatically a member of the group?

No. `gpasswd -A` only grants administration rights; add them with `gpasswd -a` if they also need the group's access.

Q16. Why is /etc/gshadow separate from /etc/group?

`/etc/group` must be readable by all tools, so sensitive data like group passwords is kept in a root-only file.

Sources (shadow-utils man pages, man7.org):
man7.org/linux/man-pages/man8/groupadd.8.html • man8/usermod.8.html • man1/gpasswd.1.html • man8/groupdel.8.html • man1/newgrp.1.html • man5/gshadow.5.html
Options can differ slightly between distributions and tool versions - check `man <command>` on your own system.